Combinatorics

Lecture 7

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Define discriminant of $\sigma \in S_n$ as $N(\sigma) = n - |C|$, where $|C|$ is the number of cycles in a disjoint cycle decomposition of σ. Note that we may not omit single element cycles now, they count towards $|C|$.

Example. In S_5 we have for example:

$$N(Id) = N((1)(2)(3)(4)(5)) = 5 - 5 = 0$$

and

$$N((134)(25)) = 5 - 2 = 3.$$

Recall that cycles of length 2 are called transpositions.

Proposition 1.

If σ is the product of k transpositions , then $N(\sigma)$ and k have the same parity.

Proof. Let σ be any permutation with k cycles in a disjoint cycle decomposition. Let (ab) be any transposition. We determine the number of cycles in a disjoint cycle decomposition of $(ab)\sigma$ depending on a and b.

• (Case 1: a and b are part of different cycles in σ.) Then there is a disjoint cycle decomposition of σ looking like this:

$$(aX)(bY)C_3 \dots C_k$$

where X and Y are sequences of elements and $C_3 \dots C_k$ are cycles. Then

$$(ab)\sigma = (ab)(aX)(bY)C_3 \dots C_k = (aYbX)C_3 \dots C_k$$

The last term is a disjoint cycle decomposition of $(ab)\sigma$ into $k-1$ cycles.

• (Case 2: a and b are part of the same cycle in σ.) Then there is a disjoint cycle decomposition of σ looking like this:

$$\sigma = (aXbY)C_2 \ldots C_k$$

where X and Y are sequences of elements and $C_2 \ldots C_k$ are cycles. Then

$$(ab)\sigma = (ab)(aXbY)C_2 \ldots C_k = (aY)(bX)C_2 \ldots C_k$$

The last term is a disjoint cycle decomposition of $(ab)\sigma$ into $k+1$ cycles.■

Stirling I

We call the number $s^I_k(n)$ of $n-$permutations of $\{1,2,\ldots,n\}$ with exactly k cycles in a disjoint cycle decomposition the unsigned Stirling number of first kind. More precisely $s^I_k(n) = |\{\sigma \in S_n | N(\sigma) = n-k\}|$.

Theorem 2.

For all $n, k \geq 1$
(1) $s^I_0(0) = 1$ and $s^I_k(0) = s^I_0(n) = 0$,
(2) $s^I_k(n) = (n-1)s^I_k(n-1) + s^I_{k-1}(n-1)$

Proof. (1) The empty permutation is the only permutation of 0 elements. Thus $s^I_0(0) = 1$ and $s^I_k(0) = 1$, since the empty permutation has no cycle and $k \geq 1$.
On the other hand every permutation of $n \geq 1$ elements has at least one cycle and hence $s^I_0(n) = 0$

(2) Let $S_n^k \subset S_n$ denote the set of permutations in S_n with exactly k cycles in a disjoint cycle decomposition. Then $|S_n^k| = s_k^I(n)$. We define a map $\phi : S_n^k \to S_{n-1}^k \cup S_{n-1}^{k-1}$ as follows: consider a permutation $\sigma \in S_n^k$ and let (AnB) denote the cycle containing n (A and B may be empty).

If $(AnB) = (n)$, remove the entire cycle (n).

Otherwise replace the cycle (AnB) with (AB). In the first case there are $k-1$ cycles in a disjoint cycle decomposition of $\phi(\sigma)$. In the latter case $\phi(\sigma)$ still has k cycles in a disjoint cycle decomposition.

Consider a disjoint cycle decomposition of some $\pi \in S_{n-1}^k \cup S_{n-1}^{k-1}$. We count how many permutations $\sigma \in S_n^k$ are mapped to π.

• (Case 1: $\pi \in S_{n-1}^{k-1}$) Then the only permutation in S_n^k mapped to π is obtained by adding the cycle (n) to π.

• (Case 2: $\pi \in S_{n-1}^{k}$) In this case we may place n between any element $x \in \{1, 2, \ldots, n-1\}$ and its successor in π (i.e., put it right behind x in the cycle containing x). Then n has different positions for distinct choices of $x \in \{1, 2, \ldots, n-1\}$. Hence $n-1$ distinct elements from S_n^k are mapped to π.
Combining these two cases we see that

$$|S_n^k| = (n-1)|S_{n-1}^k| + |S_{n-1}^{k-1}| \blacksquare$$

We call the number $s^{II}_k(n)$ of partitions of $\{1, 2, \ldots, n\}$ into k non-empty parts the Stirling Numbers of the second kind.

Some values are easy to determine:

- $s^{II}_0(0) = 1$: there is one way to partition the empty set into nonempty parts.
- $s^{II}_0(n) = 0$ for $n \geq 1$.
- $s^{II}_1(n) = 1$: Every non-empty set X can be partitioned into one non-empty set in exactly one way: $X = X$.
- $s^{II}_2(n) = 2^{n-1} - 1$: We want to partition into two non-empty parts. If we consider the parts labeled (there is a first part and a second part), then choosing the first part fully determines the second and vice versa. Every subset of $\{1, 2, \ldots, n\}$ is allowed to be the first part - except for $\varnothing$ and $\{1, 2, \ldots, n\}$. This amounts to $2^n - 2$ possibilities, however, since the parts are actually unlabeled (there is no first or second), every possibility is counted twice so we need to divide by 2.

A recursion formula for $s_k^{II}(n)$ is given as:

$$s_k^{II}(n) = ks_k^{II}(n-1) + s_{k-1}^{II}(n-1)$$

To see this, count the arrangements of the n labeled balls in unlabeled boxes with no empty box as follows:

• The ball of label n may have its own box (with no other ball in it). The number of such arrangements is equal to the number of arrangements of the remaining $n-1$ balls in $k-1$ boxes such that none of those $k-1$ boxes is empty. There are $s_{k-1}^{II}(n-1)$ of those.

• The box with the ball of label n contains another ball. Then, when removing ball n, there is still at least one ball per box. So removing ball n gets us to an arrangements of $n-1$ balls in k non-empty boxes. There are $s_k^{II}(n-1)$ of those and for each there are k possibilities where ball n could have been before removal. So there are $k \cdot s_k^{II}(n-1)$ arrangements where ball n is not alone in a box.

In total, we list the objects of interest to us:

- The unsigned Stirling numbers of the first kind $s^I_k(n)$ fulfi ll the recursion:

$$s^I_k(n) = (n-1)s^I_k(n-1) + s^I_{k-1}(n-1)$$

and count the number of $n-$permutations of $\{1, \ldots, n\}$ with k cycles.

- The signed Stirling numbers of the first kind are simply

$$\tilde{s}^I_k(n) = (-1)^{n-k} s^I_k(n)$$

- The Stirling numbers of the second kind $s^{II}_k(n)$ fulfill

$$s^{II}_k(n) = k s^{II}_k(n-1) + s^{II}_{k-1}(n-1)$$

and count the number of partitions of $\{1, \ldots, n\}$ into k non-empty parts.

Now we need some additional technique for generating functions. Sometimes it is convenient to expand generating functions in powers $\{\frac{x^n}{n!}\}$ (in particular, it will be convenient for studying Stirling numbers).

Some ideas. Say the numbers (a_n) and (b_n) count arrangements of type A and B, respectively, using n labeled objects.

Assume arrangements of type C with n objects are obtained by a unique split of the n objects into two sets and then forming an arrangement of type A with the first set and an arrangement of type B with the second.

Then c_n, the number of arrangements of type C and size n, is given as

$$c_n = \sum_{k=0}^{n} C_n^k a_k b_{n-k}$$

Crucially, the exponential generating functions $A(x), B(x), C(x)$ of the three sequences reflect this relationship as $C(x) = A(x)B(x)$ (Exercise!).

Additionally, we note that an index shift by one in the sequence corresponds to a derivation, meaning

$$F(x) = \sum_{n=0}^{\infty} a_n \frac{x^n}{n!} \Rightarrow F'(x) = \sum_{n=0}^{\infty} a_{n+1} \frac{x^n}{n!}$$

Theorem 3.

For $k \in \mathbb{N}$, we have the following identity for the exponential generating function $F_k(x)$ of the Stirling numbers $(s_k^{II}(n))_{n\in\mathbb{N}}$:

$$F_k(x) = \sum_{n=0}^{\infty} s_k^{II}(n) \frac{x^n}{n!} = \frac{1}{k!}(e^x - 1)^k$$

Proof. We proceed by induction on k. For $k = 1$ we have:

$$s_k^{II}(n) = \begin{cases} 1, & \text{for } n \geq 1; \\ 0, & \text{for } n = 0. \end{cases}$$

and the claim follows from the identity $\sum_{n=1}^{\infty} \frac{x^n}{n!} = e^x - 1$
If $k \geq 2$, we use the recursion of Stirling numbers from above. Taking the exponential generating functions for each term and then applying the induction hypothesis yields:

$$F_k'(x) = kF_k(x) + F_{k-1}(x) = kF_k(x) + \frac{1}{(k-1)!}(e^x - 1)^{k-1}$$

This differential equation has the solution $F_k(x) = \frac{1}{k!}(e^x - 1)^k$ ■

Recall that in the last lecture we defined
$p(x,k) := x(x-1)\dots(x-k+1)$ for any $x \in \mathbb{R}$.

Lemma 4.

(1) $\sum_{k=0}^{n} s_k^I(n)x^k = p(x+n-1,n)$
(2) $\sum_{k=0}^{n} \tilde{s}_k^I(n)x^k = p(x,n)$

Proof. Note that $p(x+n-1,n)$ is a polynomial of degree n with variable x so there are constants $b_{n,k}$ such that $p(x+n-1,n) = \sum_{k=0}^{n} b_{n,k}x^k$ They fulfill the following equation:

$$\sum_{k=0}^{n} b_{n,k}x^k = (x+n-1)p(x+n-2,n-1) = (x+n-1)\sum_{k=0}^{n-1} b_{n-1,k}x^k =$$

$$= \sum_{k=1}^{n} b_{n-1,k-1}x^k + (n-1)\sum_{k=0}^{n-1} b_{n-1,k}x^k$$

Comparing the coefficients of x_k we get:

$$b_{n,k} = b_{n-1,k-1} + (n-1)b_{n-1,k}$$

where we define $b_{n,k} = 0$ for $k > n$ or $k < 0$. So the numbers $b_{n,k}$ fulfill the same recursion as $s_k^I(n)$ and since the starting values $s_0^I(0) = b_{0,0} = 1$ match as well, this proves $b_{n,k} = s_k^I(n)$ and thus (1).

For (2), plug $-x$ into the equation above:

$$\sum_{k=0}^{n}(-1)^k s_k^I(n)x^k = p(-x+n-1,n) = \prod_{k=1}^{n}(-x+n-k) =$$

$$= (-1)^n \prod_{k=1}^{n}(x-n+k) = (-1)^n p(x,n)$$

Multiplying by $(-1)^n$ gives the desired result. ■

Proposition 5.

The exponential generating function of $(\tilde{s}^l_k(n))$ fulfills:

$$\sum_{n=0}^{\infty} \tilde{s}^l_k(n)\frac{x^n}{n!} = \frac{1}{k!}(log(1+x))^k$$

Proof. $(1+x)^z = e^{zlog(1+x)} = \sum_{k=0}^{\infty} \frac{1}{k!}(log(1+x))^k z^k$. On the other hand, we can also expand $(1+x)^z$ using binomial formula and Lemma 4:

$$(1+x)^z = \sum_{n=0}^{\infty} C_z^n x^n = \sum_{n=0}^{\infty} \frac{p(z,n)}{n!}x^n = \sum_{n=0}^{\infty} \frac{x^n}{n!} \sum_{k=0}^{n} \tilde{s}^l_k(n) z^k =$$

$$= \sum_{n=0}^{\infty} \sum_{k=0}^{\infty} \tilde{s}^l_k(n)\frac{x^n}{n!}z^k = \sum_{k=0}^{\infty} \left(\sum_{n=0}^{\infty} \tilde{s}^l_k(n)\frac{x^n}{n!} \right) z^k$$

We have written $(1+x)^z$ in two ways and the coefficients of z^k in both representations must match. This proves the claim. ■

Bernoulli numbers

The Bernoulli numbers B_n are defined by the following exponential generating function:

$$\sum_{n=0}^{\infty} \frac{B_n}{n!} x^n = \frac{x}{e^x - 1}$$

Since this discovery, the Bernoulli numbers have appeared in many important results.
Let us note that a recursion for the Bernoulli numbers can be found by multiplying by $e^x - 1 = \sum_{n=1}^{\infty} \frac{x^n}{n!}$ and comparing coefficients:

$$\Big(\sum_{n=1}^{\infty} \frac{x^n}{n!}\Big)\Big(\sum_{n=0}^{\infty} \frac{B_n}{n!} x^n\Big) = x$$

Thus

$$\sum_{k=1}^{n} C_n^k B_{n-k} = \begin{cases} 1, & \text{for } n = 1; \\ 0, & \text{otherwise.} \end{cases}$$

Clearly, we can rewrite the formula above (interchange k and $n-k$) as

$$\sum_{k=0}^{n-1} C_n^k B_k = \begin{cases} 1, & \text{for } n = 1; \\ 0, & \text{otherwise.} \end{cases}$$

The first few values are:
$B_0 = 1, B_1 = -\frac{1}{2}, B_2 = \frac{1}{6}, B_3 = 0, B_4 = -\frac{1}{30}, B_5 = 0.$

All odd-indexed Bernoulli numbers, except for B_1, are actually 0 (This is due to the fact that $\frac{x}{e^x-1} + \frac{x}{2}$ is an even function).

In order for us to further investigate the properties of Bernoulli numbers, let's move our attention to a generalization: the Bernoulli polynomials:

$$B_m(y) = \sum_{k=0}^{m} C_m^k B_k y^{m-k}$$

It $B_m(0) = B_m$ as well as

$$B_m(1) = \begin{cases} B_m + 1, & \text{for } m = 1; \\ B_m, & \text{otherwise.} \end{cases}$$

by recursive formula above.

Bernoulli numbers

Consider now the sum $s_m(n) = \sum\limits_{k=1}^{n} k^m$. In Lecture 2 we calculated $s_1(n), s_2(n)$ and $s_3(n)$. What about the general formula?

Theorem 6.

$$s_m(n-1) = \frac{1}{m+1}\Big(\sum_{k=0}^{m} C_{m+1}^k B_k n^{m+1-k}\Big)$$

Proof. Consider the exponential generating function of $s_m(n-1)$:

$$S(x) = \sum_{m=0}^{\infty} \frac{s_m(n-1)}{m!} x^m = \sum_{m=0}^{\infty} \sum_{k=1}^{n-1} \frac{k^m x^m}{m!}$$

Interchanging the order of summation, we find

$$S(x) = \sum_{k=1}^{n-1} \sum_{m=0}^{\infty} \frac{k^m x^m}{m!} = \sum_{k=1}^{n-1} e^{kx} = \frac{e^{nx} - e^x}{e^x - 1}$$

Compare this to the exponential generating function for Bernoulli polynomials:

$$\sum_{m=0}^{\infty} \frac{B_m(y)}{m!} x^m = \sum_{m=0}^{\infty} \sum_{k=0}^{m} \frac{1}{m!} C_m^k B_k y^{m-k} x^m =$$

$$= \sum_{k=0}^{\infty} \sum_{m=k}^{\infty} \frac{1}{k!(m-k)!} B_k y^{m-k} x^m = \sum_{k=0}^{\infty} \frac{B_k x^k}{k!} \sum_{m=k}^{\infty} \frac{1}{(m-k)!} y^{m-k} x^{m-k} =$$

$$= \sum_{k=0}^{\infty} \frac{B_k x^k}{k!} \sum_{m=0}^{\infty} \frac{1}{m!} y^m x^m = \sum_{k=0}^{\infty} \frac{B_k x^k}{k!} e^{xy} = \frac{x e^{xy}}{e^x - 1}$$

Now we can extract the mth coefficient from $S(x)$ (exercise!):

$$s_m(n-1) = \frac{B_{m+1}(n) - B_{m+1}}{m+1}$$

which proves the theorem.■

Example. For $m = 5$, we find the formula

$$s_5(n) = n^5 + s_5(n-1) = n^5 + \frac{1}{6}\Big(\sum_{k=0}^{5} C_6^k B_k n^{6-k}\Big) =$$

$$= n^5 + \frac{1}{6}\Big(n^6 - 3n^5 + \frac{5}{2}n^4 - \frac{1}{2}n^2\Big) = \frac{n^2(n+1)^2(2n^2+2n-1)}{12}$$